

(12) **United States Patent**
Birch et al.

(10) **Patent No.:** **US 12,407,684 B2**
(45) **Date of Patent:** **Sep. 2, 2025**

(54) **SYSTEM AND METHOD FOR
IMPLEMENTING AUTHENTICATION
CONTROL PROTOCOLS VIA COMPONENTS
OF AN ENTITY DEVICE**

(58) **Field of Classification Search**
CPC H04L 63/0861; G06V 40/161;
G06V 40/168; G06V 40/172
See application file for complete search history.

(71) Applicant: **BANK OF AMERICA
CORPORATION**, Charlotte, NC (US)

(56) **References Cited**
U.S. PATENT DOCUMENTS

(72) Inventors: **Christopher Daniel Birch**, Tega Cay,
SC (US); **Sophie Morgan Danielpour**,
Durham, NC (US); **Susan R. Hart**,
Addison, TX (US); **Son Huynh**,
Charlotte, NC (US); **Tyler Keith**
Prothro, Waxahachie, TX (US);
Shannon Sabina Rolinger, Pineville,
NC (US); **Amado Jose Martinez**
Suarez, Addison, TX (US); **Stephen**
Jack Williams, Charlotte, NC (US)

10,055,723 B2	8/2018	Buchholtz et al.
10,318,954 B1	6/2019	Gailloux et al.
10,332,214 B2	6/2019	Freeman
10,580,049 B2	3/2020	Graylin
10,762,579 B2	9/2020	Riechers et al.
11,107,056 B2	8/2021	Lee
11,132,689 B2	9/2021	Riechers et al.
11,195,182 B2	12/2021	Ozvat et al.
11,222,330 B2	1/2022	Dua

(Continued)

Primary Examiner — Carl G Colin

Assistant Examiner — Andrew Suh

(74) *Attorney, Agent, or Firm* — Moore & Van Allen
PLLC; Anoohya Yarlagadda

(73) Assignee: **BANK OF AMERICA
CORPORATION**, Charlotte, NC (US)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 295 days.

(21) Appl. No.: **17/984,919**

(22) Filed: **Nov. 10, 2022**

(65) **Prior Publication Data**
US 2024/0163283 A1 May 16, 2024

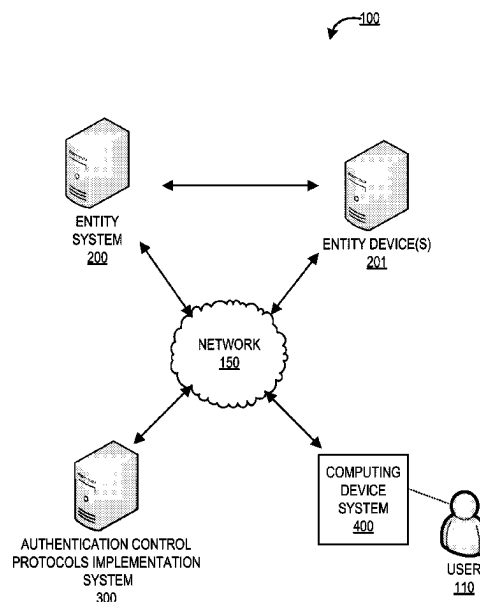
(51) **Int. Cl.**
H04L 9/40 (2022.01)
G06V 40/16 (2022.01)

(52) **U.S. Cl.**
CPC **H04L 63/0861** (2013.01); **G06V 40/161**
(2022.01); **G06V 40/168** (2022.01)

(57) **ABSTRACT**

Embodiments of the present invention provide a system for implementing authentication control protocols via components of entity device. In particular, the system may be configured to determine initiation of a resource interaction by a user at an entity device, control one or more components of the entity device to cause the entity device to capture a real-time image of the user, compare the real-time image captured via the entity device with an updated enhanced user image stored in a data file of the user, in response to comparing the real-time image with the updated enhanced user image, determine if the real-time image matches the updated enhanced user image, and implement one or more authentication control protocols based on determining if the real-time image matches the updated enhanced user image.

20 Claims, 5 Drawing Sheets



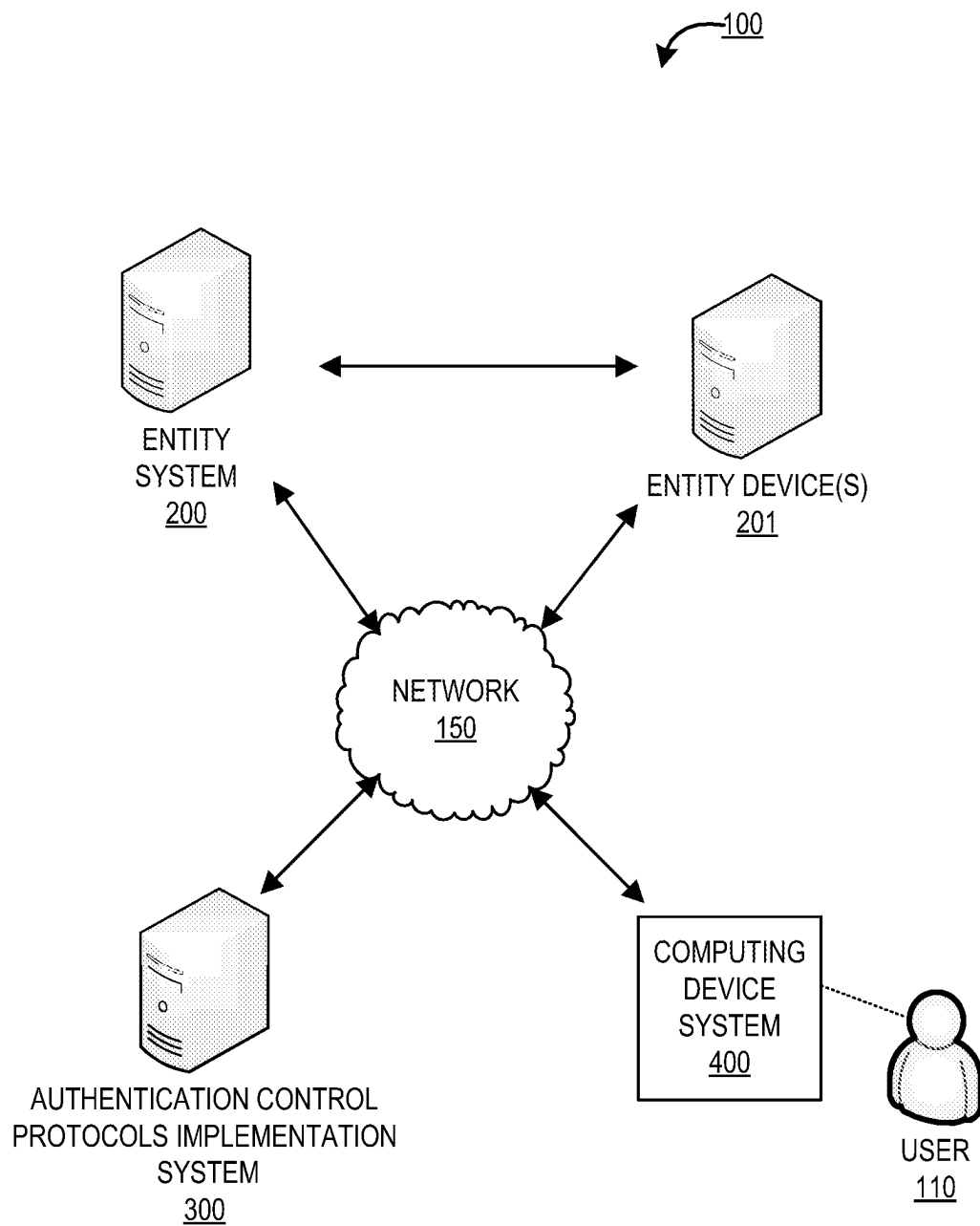
(56)

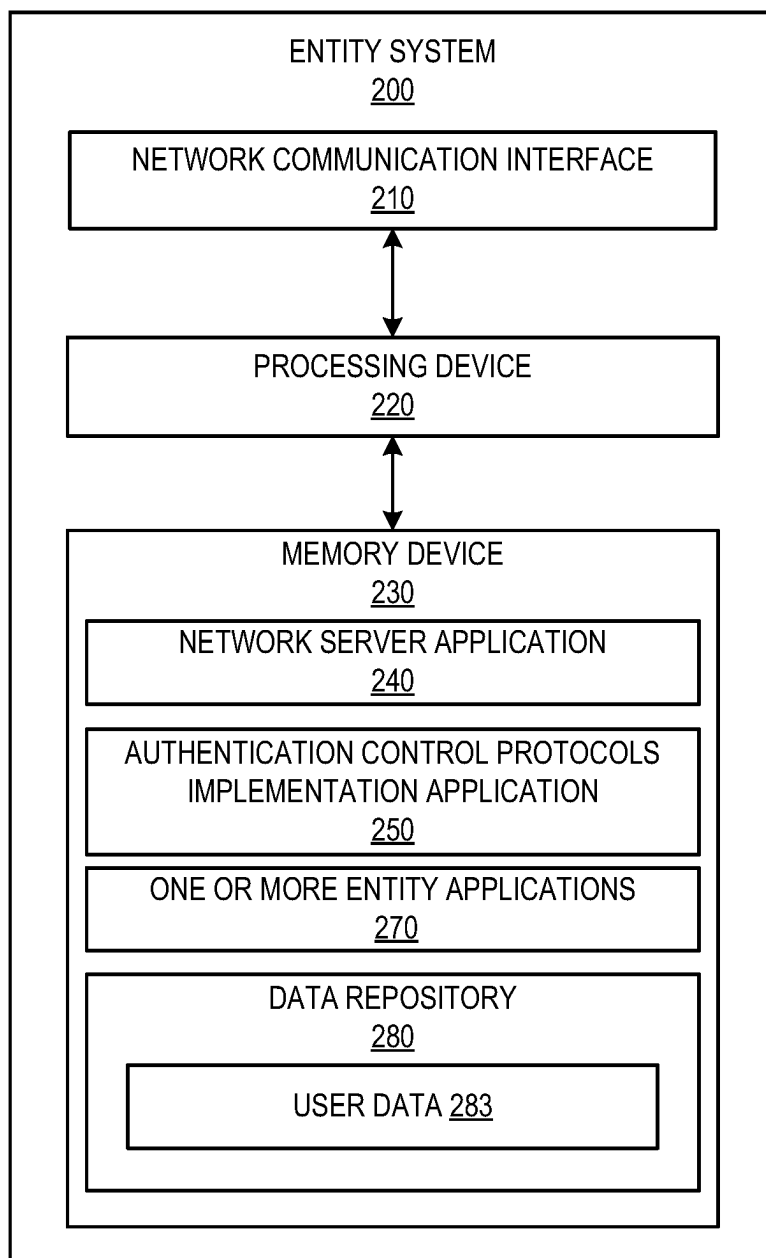
References Cited

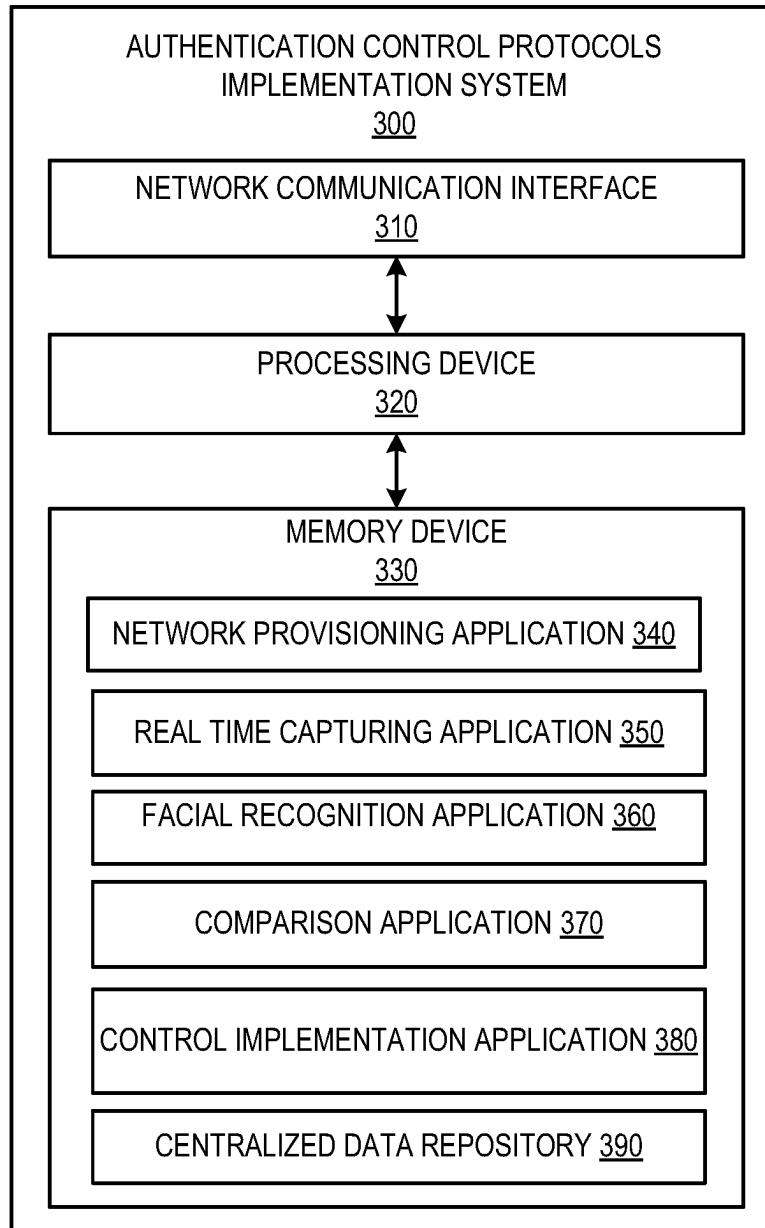
U.S. PATENT DOCUMENTS

11,227,284	B2	1/2022	Safak	
11,301,840	B1	4/2022	Smith et al.	
11,301,841	B2	4/2022	Ko et al.	
2010/0125510	A1	5/2010	Monahan et al.	
2014/0283014	A1 *	9/2014	Tse	G06F 21/316 726/19
2018/0075438	A1 *	3/2018	Iqbal	G06Q 20/3223
2018/0150816	A1	5/2018	Liu et al.	
2018/0293562	A1	10/2018	Squire et al.	
2019/0034900	A1	1/2019	Lo et al.	
2019/0147156	A1 *	5/2019	Burri	G06V 40/174 713/186
2020/0005262	A1 *	1/2020	Arora	G06Q 20/3224
2021/0314526	A1 *	10/2021	Astarabadi	G06V 20/47
2022/0051245	A1	2/2022	Riechers et al.	
2022/0253843	A1	8/2022	Gleeson et al.	
2022/0392267	A1 *	12/2022	Li	G06V 40/50
2023/0073410	A1 *	3/2023	Tussy	G06F 21/32

* cited by examiner

**FIG. 1**

**FIG. 2**

**FIG. 3**

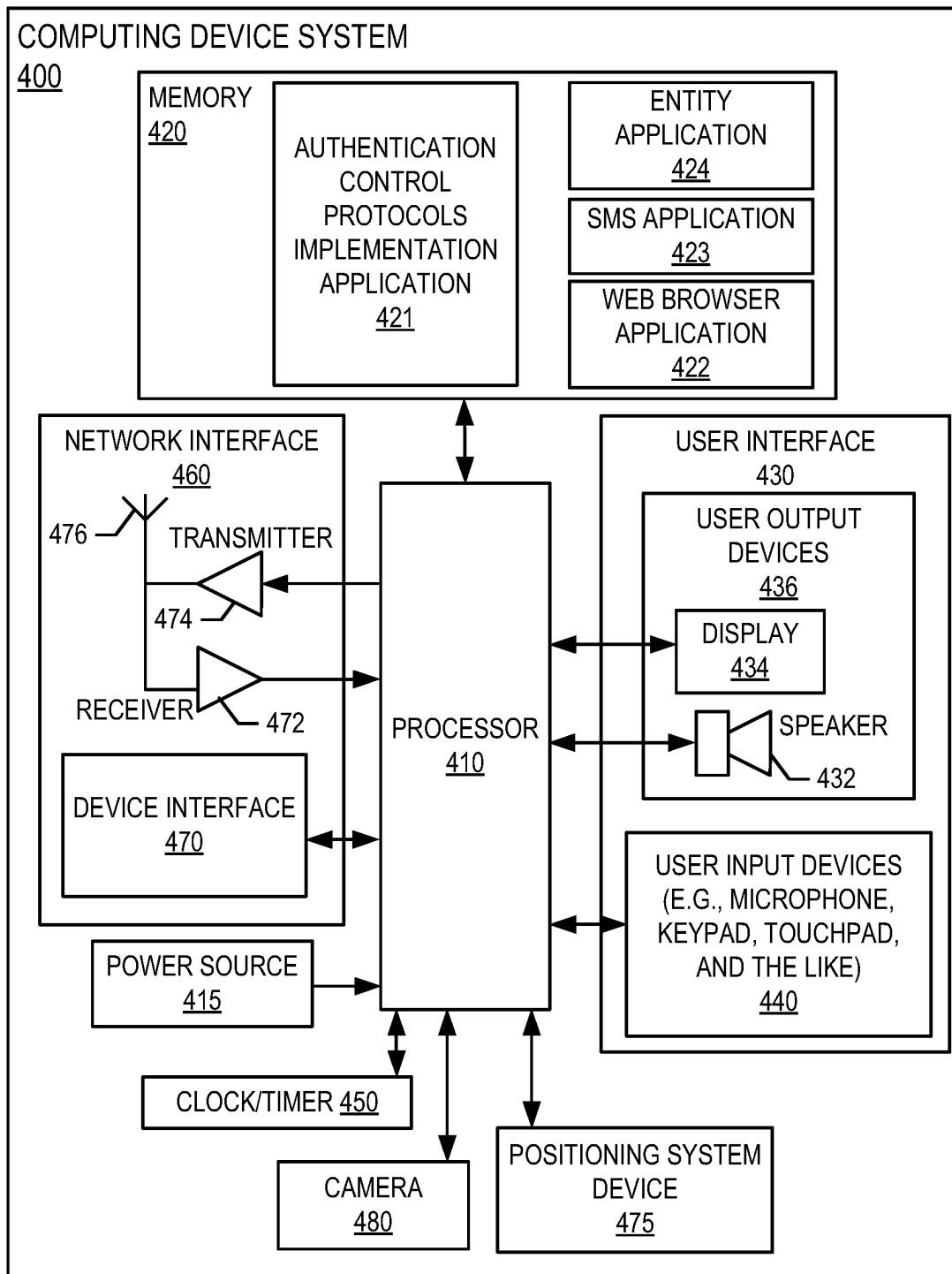
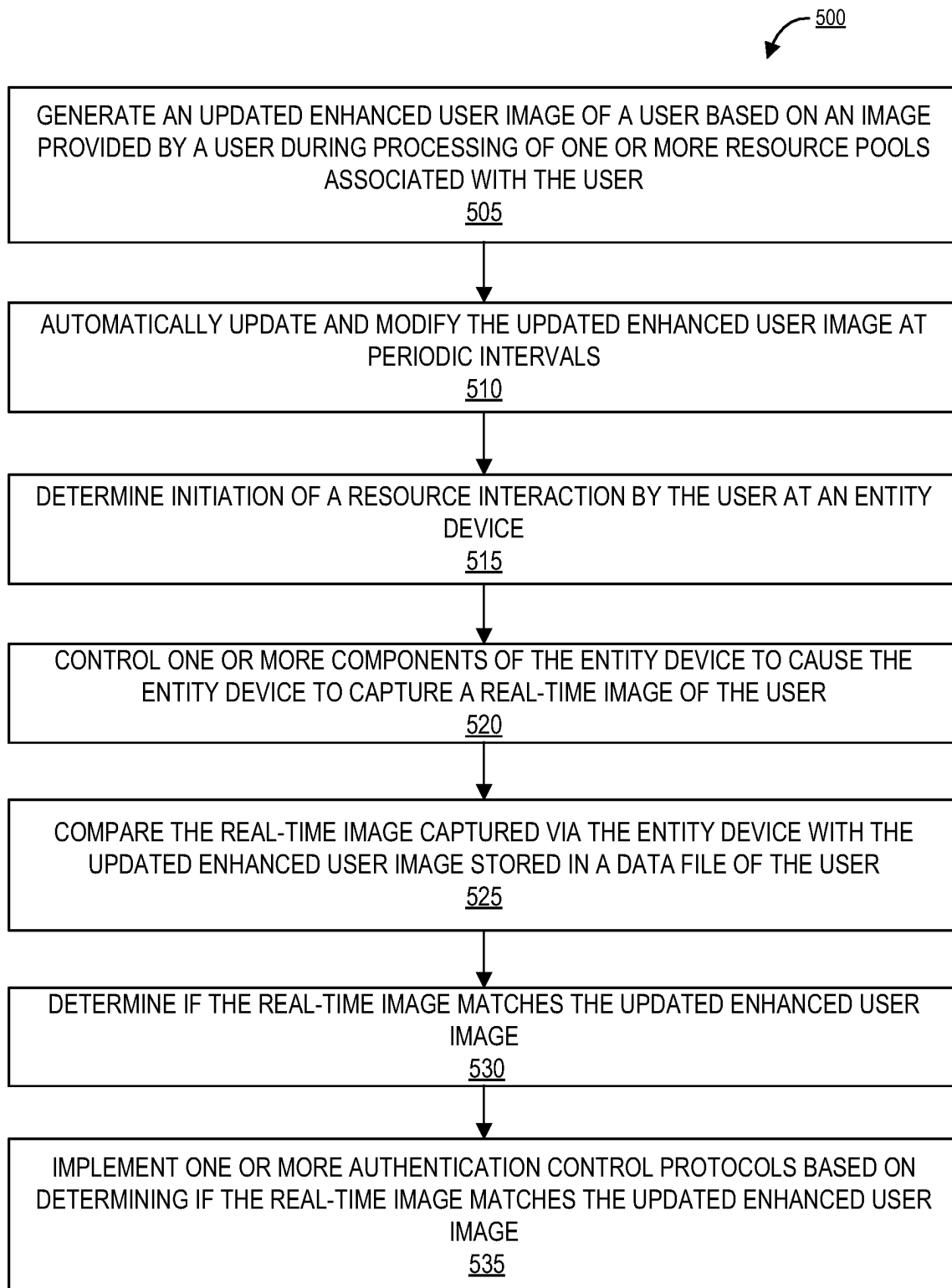


FIG. 4

**FIG. 5**

1

SYSTEM AND METHOD FOR IMPLEMENTING AUTHENTICATION CONTROL PROTOCOLS VIA COMPONENTS OF AN ENTITY DEVICE

BACKGROUND

There exists a need for a system to implement authentication control protocols via components of entity devices.

BRIEF SUMMARY

The following presents a summary of certain embodiments of the invention. This summary is not intended to identify key or critical elements of all embodiments nor delineate the scope of any or all embodiments. Its sole purpose is to present certain concepts and elements of one or more embodiments in a summary form as a prelude to the more detailed description that follows.

Embodiments of the present invention address the above needs and/or achieve other advantages by providing apparatuses (e.g., a system, computer program product and/or other devices) and methods for implementing authentication control protocols via components of entity device. The system embodiments may comprise one or more memory devices having computer readable program code stored thereon, a communication device, and one or more processing devices operatively coupled to the one or more memory devices, wherein the one or more processing devices are configured to execute the computer readable program code to carry out the invention. In computer program product embodiments of the invention, the computer program product comprises at least one non-transitory computer readable medium comprising computer readable instructions for carrying out the invention. Computer implemented method embodiments of the invention may comprise providing a computing system comprising a computer processing device and a non-transitory computer readable medium, where the computer readable medium comprises configured computer program instruction code, such that when said instruction code is operated by said computer processing device, said computer processing device performs certain operations to carry out the invention.

In some embodiments, the present invention determines initiation of a resource interaction by a user at an entity device, controls one or more components of the entity device to cause the entity device to capture a real-time image of the user, compares the real-time image captured via the entity device with an updated enhanced user image stored in a data file of the user, in response to comparing the real-time image with the updated enhanced user image, determines if the real-time image matches the updated enhanced user image, and implements one or more authentication control protocols based on determining if the real-time image matches the updated enhanced user image.

In some embodiments, the present invention determines that the real-time image matches the updated enhanced user image and implement the one or more authentication control protocols comprising a complete access protocol to allow complete access to one or more resource pools of the user to complete the resource interaction.

In some embodiments, the present invention determines that the real-time image does not match the updated enhanced user image and implement the one or more authentication control protocols comprising at least one of a limited access protocol to provide limited access to one or more resource pools of the user, a restricting access protocol to

2

restrict access to the one or more resource pools of the user, and a blocking protocol to block the resource interaction initiated by the user.

In some embodiments, the present invention implements the limited access protocol based on an exposure protocol selected by the user.

In some embodiments, the present invention performs one or more actions in response to determining that the real-time image does not match the updated enhanced user image, wherein the one or more actions comprise increasing an exposure rating associated with the user, transmitting one or more alerts to a register user device of the user, locking the one or more resource pools of the user based on determining a number of mismatches associated with comparison of the real-time image and the updated enhanced user image is greater than a predetermined limit, and transmitting an exposure data file of the user to one or more entity partners.

In some embodiments, the present invention generates the updated enhanced user image based on an image provided by the user during processing of the one or more resource pools.

In some embodiments, the present invention modifies the updated enhanced user image at periodic intervals, wherein updating the enhanced user image comprises computing changes associated with progressive appearance of the user and modifying the updated enhanced user image generated based on the image provided by the user to reflect the computed changes.

The features, functions, and advantages that have been discussed may be achieved independently in various embodiments of the present invention or may be combined with yet other embodiments, further details of which can be seen with reference to the following description and drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

Having thus described embodiments of the invention in general terms, reference will now be made to the accompanying drawings, wherein:

FIG. 1 provides a block diagram illustrating a system environment for implementing authentication control protocols via components of entity device, in accordance with an embodiment of the invention;

FIG. 2 provides a block diagram illustrating the entity system 200 of FIG. 1, in accordance with an embodiment of the invention;

FIG. 3 provides a block diagram illustrating an authentication control protocols implementation system 300 of FIG. 1, in accordance with an embodiment of the invention;

FIG. 4 provides a block diagram illustrating the computing device system 400 of FIG. 1, in accordance with an embodiment of the invention;

FIG. 5 provides a block diagram illustrating a process flow for implementing authentication control protocols via components of entity device, in accordance with an embodiment of the invention.

DETAILED DESCRIPTION OF EMBODIMENTS OF THE INVENTION

Embodiments of the present invention will now be described more fully hereinafter with reference to the accompanying drawings, in which some, but not all, embodiments of the invention are shown. Indeed, the invention may be embodied in many different forms and should not be construed as limited to the embodiments set forth

3

herein; rather, these embodiments are provided so that this disclosure will satisfy applicable legal requirements. Where possible, any terms expressed in the singular form herein are meant to also include the plural form and vice versa, unless explicitly stated otherwise. Also, as used herein, the term “a” and/or “an” shall mean “one or more,” even though the phrase “one or more” is also used herein. Furthermore, when it is said herein that something is “based on” something else, it may be based on one or more other things as well. In other words, unless expressly indicated otherwise, as used herein “based on” means “based at least in part on” or “based at least partially on.” Like numbers refer to like elements throughout.

As used herein, the term “resource entity” or “entity” may be any institution which involves in financial transactions. In one embodiment, the term “entity” or “resource entity” may be any financial institution. As used herein, the term “entity device” may be any device associated with the entity. In some embodiments of the present invention, the entity device may be an automated machine. In some embodiments, the “automated machine” may be any automated device that is involved in distribution of resources such as cash, checks, electronic transfers, money orders or the like which may be performed using a credit card, a debit card, or the like. In one embodiment of the present invention, the automated machine may be an Automated Teller Machine (ATM). In some embodiments, the “automated machine” may be any automated device that provides information associated with resource pools of the user and/or allows the user to perform resource interactions (e.g., transactions). In some embodiments, the automated machine may be owned and/or controlled by the resource entity.

As described herein, a “user” may be a customer or a potential customer of the entity. In some embodiments, a “user” may be a financial institution customer (e.g., an account holder or a person who has an account (e.g., banking account, credit account, or the like)). An “account” or “resource pool” may be the relationship that the customer has with the financial institution. Examples of accounts include a deposit account, such as a transactional account (e.g. a banking account), a savings account, an investment account, a money market account, a time deposit, a demand deposit, a pre-paid account, a credit account, a non-monetary customer information that includes only personal information associated with the customer, or the like. The account is associated with and/or maintained by a financial institution.

A “user interface” is any device or software that allows a user to input information, such as commands or data, into a device, or that allows the device to output information to the user. For example, the user interface includes a graphical user interface (GUI) or an interface to input computer-executable instructions that direct a processing device to carry out specific functions. The user interface typically employs certain input and output devices to input data received from a user or to output data to a user. These input and output devices may include a display, mouse, keyboard, button, touchpad, touch screen, microphone, speaker, LED, light, joystick, switch, buzzer, bell, and/or other user input/output device for communicating with one or more users.

Conventional systems do not have the capability to implement authentication control protocols instantaneously to prevent misappropriation of resources associated with users of an entity. As such, there exists a need for a system that instantaneously implements authentication control protocols via one or more components of entity devices provided by the entity. The system of the present invention solves this problem by embedding facial recognition technology into

4

entity devices to implement authentication control protocols on resource interactions initiated by users via the entity devices.

FIG. 1 provides a block diagram illustrating a system environment **100** for implementing authentication control protocols via components of entity device, in accordance with an embodiment of the invention. As illustrated in FIG. 1, the environment **100** includes an authentication control protocols implementation system **300**, entity system **200**, one or more entity devices **201**, and a computing device system **400**. One or more users **110** may be included in the system environment **100**, where the users **110** interact with the other entities of the system environment **100** via a user interface of the computing device system **400**. In some embodiments, the one or more user(s) **110** of the system environment **100** may be customers of an entity associated with the entity system **200**.

The entity system(s) **200** may be any system owned or otherwise controlled by an entity to support or perform one or more process steps described herein. In some embodiments, the entity is a financial institution. In some embodiments, the one or more entity devices **201** may be owned, operated, controlled, and/or maintained by the entity. Examples of the one or more entity devices may include, but are not limited to, Automated Teller Machines, automated kiosks, associate computing devices, or the like that allow a user to access information associated with one or more resource pools of the user and/or to perform one or more resource interactions associated with the one or more resource pools of the user. In a preferred embodiment of the present invention, the entity device **200** may be an Automated Teller Machine.

The authentication control protocols implementation system **300** is a system of the present invention for performing one or more process steps described herein. In some embodiments, the authentication control protocols implementation system **300** may be an independent system. In some embodiments, the authentication control protocols implementation system **300** may be a part of the entity system **200**.

The authentication control protocols implementation system **300**, the entity system **200**, and the computing device system **400** may be in network communication across the system environment **100** through the network **150**. The network **150** may include a local area network (LAN), a wide area network (WAN), and/or a global area network (GAN). The network **150** may provide for wireline, wireless, or a combination of wireline and wireless communication between devices in the network. In one embodiment, the network **150** includes the Internet. In general, the authentication control protocols implementation system **300** is configured to communicate information or instructions with the entity system **200**, and/or the computing device system **400** across the network **150**.

The computing device system **400** may be a system owned or controlled by the entity of the entity system **200** and/or the user **110**. As such, the computing device system **400** may be a computing device of the user **110**. In general, the computing device system **400** communicates with the user **110** via a user interface of the computing device system **400**, and in turn is configured to communicate information or instructions with the authentication control protocols implementation system **300**, and/or entity system **200** across the network **150**. In some embodiments of the invention, the computing device system **400** may be a mobile device.

FIG. 2 provides a block diagram illustrating the entity system **200**, in greater detail, in accordance with embodiments of the invention. As illustrated in FIG. 2, in one

5

embodiment of the invention, the entity system **200** includes one or more processing devices **220** operatively coupled to a network communication interface **210** and a memory device **230**. In certain embodiments, the entity system **200** is operated by a first entity, such as a financial institution, while in other embodiments, the entity system **200** is operated by an entity other than a financial institution.

It should be understood that the memory device **230** may include one or more databases or other data structures/repositories. The memory device **230** also includes computer-executable program code that instructs the processing device **220** to operate the network communication interface **210** to perform certain communication functions of the entity system **200** described herein. For example, in one embodiment of the entity system **200**, the memory device **230** includes, but is not limited to, an authentication control protocols implementation application **250**, one or more entity applications **270**, and a data repository **280** comprising user data **283**, where the user data **283** comprises data associated with users **110** and/or one or more resource pools associated with the users **110**. The one or more entity applications **270** may be any applications provided by the entity that allows users **110** to access information and/or perform one or more actions associated with one or more resource pools of the users **110**. The computer-executable program code of the network server application **240**, the authentication control protocols implementation application **250**, the one or more entity application **270** to perform certain logic, data-extraction, and data-storing functions of the entity system **200** described herein, as well as communication functions of the entity system **200**.

The network server application **240**, the authentication control protocols implementation application **250**, and the one or more entity applications **270** are configured to store data in the data repository **280** or to use the data stored in the data repository **280** when communicating through the network communication interface **210** with the authentication control protocols implementation system **300**, and/or the computing device system **400** to perform one or more process steps described herein. In some embodiments, the entity system **200** may receive instructions from the authentication control protocols implementation system **300** via the authentication control protocols implementation application **250** to perform certain operations. The authentication control protocols implementation application **250** may be provided by the authentication control protocols implementation system **300**.

FIG. 3 provides a block diagram illustrating the authentication control protocols implementation system **300** in greater detail, in accordance with embodiments of the invention. As illustrated in FIG. 3, in one embodiment of the invention, the authentication control protocols implementation system **300** includes one or more processing devices **320** operatively coupled to a network communication interface **310** and a memory device **330**. In certain embodiments, the authentication control protocols implementation system **300** is operated by a first entity, such as a financial institution, while in other embodiments, the authentication control protocols implementation system **300** is operated by an entity other than a financial institution. In some embodiments, the authentication control protocols implementation system **300** is owned or operated by the entity of the entity system **200**. In some embodiments, the authentication control protocols implementation system **300** may be an independent system. In alternate embodiments, the authentication control protocols implementation system **300** may be a part of the entity system **200**.

6

It should be understood that the memory device **330** may include one or more databases or other data structures/repositories. The memory device **330** also includes computer-executable program code that instructs the processing device **320** to operate the network communication interface **310** to perform certain communication functions of the authentication control protocols implementation system **300** described herein. For example, in one embodiment of the authentication control protocols implementation system **300**, the memory device **330** includes, but is not limited to, a network provisioning application **340**, a real-time capturing application **350**, a facial recognition application **360**, a comparison application **370**, a control implementation application **380**, and a centralized data repository **390** comprising data processed or accessed by one or more applications in the memory device **330**. The computer-executable program code of the network provisioning application **340**, the real-time capturing application **350**, the facial recognition application **360**, the comparison application **370**, and the control implementation application **380** may instruct the processing device **320** to perform certain logic, data-processing, and data-storing functions of the authentication control protocols implementation system **300** described herein, as well as communication functions of the authentication control protocols implementation system **300**.

The network provisioning application **340**, the real-time capturing application **350**, the facial recognition application **360**, the comparison application **370**, and the control implementation application **380** are configured to invoke or use the data in the data repository **390** when communicating through the network communication interface **310** with the entity system **200**, and/or the computing device system **400**. In some embodiments, the network provisioning application **340**, the real-time capturing application **350**, the facial recognition application **360**, the comparison application **370**, and the control implementation application **380** may store the data extracted or received from the entity system **200**, and the computing device system **400** in the centralized data repository **390**. In some embodiments, the network provisioning application **340**, the real-time capturing application **350**, the facial recognition application **360**, the comparison application **370**, and the control implementation application **380** may be a part of a single application. The functionalities of the network provisioning application **340**, the real-time capturing application **350**, the facial recognition application **360**, the comparison application **370**, and the control implementation application **380** are explained in greater detail in FIG. 5.

FIG. 4 provides a block diagram illustrating a computing device system **400** of FIG. 1 in more detail, in accordance with embodiments of the invention. However, it should be understood that a mobile telephone is merely illustrative of one type of computing device system **400** that may benefit from, employ, or otherwise be involved with embodiments of the present invention and, therefore, should not be taken to limit the scope of embodiments of the present invention. Other types of computing devices may include portable digital assistants (PDAs), pagers, mobile televisions, cameras, video recorders, audio/video player, radio, GPS devices, wearable devices, Internet-of-things devices, augmented reality devices, virtual reality devices, automated teller machine devices, electronic kiosk devices, or any combination of the aforementioned.

Some embodiments of the computing device system **400** include a processor **410** communicably coupled to such devices as a memory **420**, user output devices **436**, user input devices **440**, a network interface **460**, a power source

415, a clock or other timer 450, a camera 480, and a positioning system device 475. The processor 410, and other processors described herein, generally include circuitry for implementing communication and/or logic functions of the computing device system 400. For example, the processor 410 may include a digital signal processor device, a micro-processor device, and various analog to digital converters, digital to analog converters, and/or other support circuits. Control and signal processing functions of the computing device system 400 are allocated between these devices according to their respective capabilities. The processor 410 thus may also include the functionality to encode and interleave messages and data prior to modulation and transmission. The processor 410 can additionally include an internal data modem. Further, the processor 410 may include functionality to operate one or more software programs, which may be stored in the memory 420. For example, the processor 410 may be capable of operating a connectivity program, such as a web browser application 422. The web browser application 422 may then allow the computing device system 400 to transmit and receive web content, such as, for example, location-based content and/or other web page content, according to a Wireless Application Protocol (WAP), Hypertext Transfer Protocol (HTTP), and/or the like.

The processor 410 is configured to use the network interface 460 to communicate with one or more other devices on the network 150. In this regard, the network interface 460 includes an antenna 476 operatively coupled to a transmitter 474 and a receiver 472 (together a “transceiver”). The processor 410 is configured to provide signals to and receive signals from the transmitter 474 and receiver 472, respectively. The signals may include signaling information in accordance with the air interface standard of the applicable cellular system of the wireless network 152. In this regard, the computing device system 400 may be configured to operate with one or more air interface standards, communication protocols, modulation types, and access types. By way of illustration, the computing device system 400 may be configured to operate in accordance with any of a number of first, second, third, and/or fourth-generation communication protocols and/or the like.

As described above, the computing device system 400 has a user interface that is, like other user interfaces described herein, made up of user output devices 436 and/or user input devices 440. The user output devices 436 include a display 430 (e.g., a liquid crystal display or the like) and a speaker 432 or other audio device, which are operatively coupled to the processor 410.

The user input devices 440, which allow the computing device system 400 to receive data from a user such as the user 110, may include any of a number of devices allowing the computing device system 400 to receive data from the user 110, such as a keypad, keyboard, touch-screen, touchpad, microphone, mouse, joystick, other pointer device, button, soft key, and/or other input device(s). The user interface may also include a camera 480, such as a digital camera.

The computing device system 400 may also include a positioning system device 475 that is configured to be used by a positioning system to determine a location of the computing device system 400. For example, the positioning system device 475 may include a GPS transceiver. In some embodiments, the positioning system device 475 is at least partially made up of the antenna 476, transmitter 474, and receiver 472 described above. For example, in one embodiment, triangulation of cellular signals may be used to

identify the approximate or exact geographical location of the computing device system 400. In other embodiments, the positioning system device 475 includes a proximity sensor or transmitter, such as an RFID tag, that can sense or be sensed by devices known to be located proximate a merchant or other location to determine that the computing device system 400 is located proximate these known devices.

The computing device system 400 further includes a power source 415, such as a battery, for powering various circuits and other devices that are used to operate the computing device system 400. Embodiments of the computing device system 400 may also include a clock or other timer 450 configured to determine and, in some cases, communicate actual or relative time to the processor 410 or one or more other devices.

The computing device system 400 also includes a memory 420 operatively coupled to the processor 410. As used herein, memory includes any computer readable medium (as defined herein below) configured to store data, code, or other information. The memory 420 may include volatile memory, such as volatile Random Access Memory (RAM) including a cache area for the temporary storage of data. The memory 420 may also include non-volatile memory, which can be embedded and/or may be removable. The non-volatile memory can additionally or alternatively include an electrically erasable programmable read-only memory (EEPROM), flash memory or the like.

The memory 420 can store any of a number of applications which comprise computer-executable instructions/code executed by the processor 410 to implement the functions of the computing device system 400 and/or one or more of the process/method steps described herein. For example, the memory 420 may include such applications as a conventional web browser application 422, an authentication control protocols implementation application 421, entity application 424, an SMS application 423, or the like. In some embodiments, the entity application 424 may be an online banking application. The authentication control protocols implementation application 421 may comprise instructions to a graphical user interface (GUI) on the display 430 that allows the user 110 to interact with the entity system 200, the authentication control protocols implementation system 300, and/or other devices or systems. The memory 420 of the computing device system 400 may comprise a Short Message Service (SMS) application 423 configured to send, receive, and store data, information, communications, alerts, and the like via the wireless telephone network 152. In some embodiments, the authentication control protocols implementation application 421 provided by the authentication control protocols implementation system 300 allows the user 110 to access the authentication control protocols implementation system 300. In some embodiments, the entity application 424 provided by the entity system 200 and the authentication control protocols implementation application 421 allow the user 110 to access the functionalities provided by the authentication control protocols implementation system 300 and the entity system 200. In some embodiments, the entity application 424 may be an online banking application.

The memory 420 can also store any of a number of pieces of information, and data, used by the computing device system 400 and the applications and devices that make up the computing device system 400 or are in communication with the computing device system 400 to implement the functions of the computing device system 400 and/or the other systems described herein.

FIG. 5 provides a block diagram illustrating a process flow 500 for implementing authentication control protocols via components of entity device, in accordance with an embodiment of the invention.

As shown in block 505, the system generates an updated enhanced user image of a user based on an image provided by a user during processing of one or more resource pools associated with the user. In some embodiments, the user may be a customer of an entity that manages the one or more resource pools of the user. The image provided by the user during processing of the one or more resource pools may be any image of the user provided or captured during the opening process associated with the one or more resources pools (e.g., credit account, savings account, checking account, or the like). In some embodiments, the system may use advanced appearance progressing technology to generate the enhanced user image of the user. For example, if the image provided by the user dates back to ten years, the system may use the advanced appearance progressing technology to generate a current up-to-date image that reflects appearance changes of the user for the last ten years.

As shown in block 510, the system automatically updates and modifies the updated enhanced user image at periodic intervals. Modifying the updated enhanced user image comprises computing changes associated with progressive appearance of the user and modifying the updated enhanced user image generated based on the image provided by the user to account for the computed changes. The system may store the updated the enhanced user image in a data file associated with the user.

As shown in block 515, the system determines initiation of a resource interaction by the user at an entity device. The resource interaction may be any type of interaction including but not limited to transfer of resources, withdrawal of resources, resource information request associated with the one or more resource pools of the user. Entity device may be Automated Teller Machines, automated kiosks, associate computing devices, or the like that allow a user to access information associated with one or more resource pools of the user and/or to perform one or more resource interactions associated with the one or more resource pools of the user. In a preferred embodiment of the present invention, the entity device may be an Automated Teller Machine.

As shown in block 520, the system controls one or more components of the entity device to cause the entity device to capture a real-time image of the user. The system may transmit one or more control signals to the entity device to cause the one or more components (e.g., software, camera, video recorder, or the like) to capture the real-time image of the user. In some embodiments, the system may package one or more software codes associated with implementation of facial recognition technology and may embed it into the one or more components of the entity device. In some embodiments, the one or more software codes may cause the entity device to automatically initiate facial recognition technology to capture the real-time image of the user.

As shown in block 525, the system compares the real-time image captured via the entity device with the updated enhanced user image stored in a data file of the user. The system may utilize facial recognition comparison technology to compare the real-time image and the updated enhanced user image. The facial recognition technology utilized by the system may be based on one or more facial recognition algorithms/methods comprising Convolutional Neural Network (CNN), Eigenfaces, Fisherfaces, PCA and SVM Kernel Methods, Haar Cascades Three-Dimensional Recognition, Skin Texture Analysis, Thermal Cameras,

adaptive neuro-fuzzy interference system (ANFIS), Local Binary Patterns Histograms (LBPH), FaceNet, NEC technology, Megvii (Face++), and/or the like.

As shown in block 530, the system determines if the real-time image matches the updated enhanced user image. As shown in block 535, the system implements one or more authentication control protocols based on determining if the real-time image matches the updated enhanced user image.

In some embodiments, the system determines that the real-time image matches the updated enhanced user image and implement the one or more authentication control protocols comprising a complete access protocol to allow complete access to one or more resource pools of the user to complete the resource interaction.

In some embodiments, the system determines that the real-time image does not match the updated enhanced user image and implements the one or more authentication control protocols comprising at least one of a limited access protocol to provide limited access to one or more resource pools of the user, a restricting access protocol to block access to the one or more resource pools of the user, and a blocking protocol to block the resource interaction initiated by the user. In some embodiments, the limited access protocol may be implemented based on an exposure protocol selected by the user. For example, if the user selects a medium exposure protocol and the medium exposure protocol allows processing of resource interactions up to a predefined value, the system may allow partial completion of the resource interaction based on the total amount of resources associated with the resource interaction.

In some embodiments, the system performs one or more actions in response to determining that the real-time image does not match the updated enhanced user image, wherein the one or more actions comprise increasing an exposure rating associated with the user, transmitting one or more alerts to a register user device of the user, locking the one or more resource pools of the user based on determining a number of mismatches associated with comparison of the real-time image and the updated enhanced user image is greater than a predetermined limit, and transmitting an exposure data file of the user to one or more entity partners (e.g., other entities, legal partners, or the like).

As will be appreciated by one of skill in the art, the present invention may be embodied as a method (including, for example, a computer-implemented process, a business process, and/or any other process), apparatus (including, for example, a system, machine, device, computer program product, and/or the like), or a combination of the foregoing. Accordingly, embodiments of the present invention may take the form of an entirely hardware embodiment, an entirely software embodiment (including firmware, resident software, micro-code, and the like), or an embodiment combining software and hardware aspects that may generally be referred to herein as a "system." Furthermore, embodiments of the present invention may take the form of a computer program product on a computer-readable medium having computer-executable program code embodied in the medium.

Any suitable transitory or non-transitory computer readable medium may be utilized. The computer readable medium may be, for example but not limited to, an electronic, magnetic, optical, electromagnetic, infrared, or semiconductor system, apparatus, or device. More specific examples of the computer readable medium include, but are not limited to, the following: an electrical connection having one or more wires; a tangible storage medium such as a portable computer diskette, a hard disk, a random access

11

memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), a compact disc read-only memory (CD-ROM), or other optical or magnetic storage device.

In the context of this document, a computer readable medium may be any medium that can contain, store, communicate, or transport the program for use by or in connection with the instruction execution system, apparatus, or device. The computer usable program code may be transmitted using any appropriate medium, including but not limited to the Internet, wireline, optical fiber cable, radio frequency (RF) signals, or other mediums.

Computer-executable program code for carrying out operations of embodiments of the present invention may be written in an object oriented, scripted or unscripted programming language such as Java, Perl, Smalltalk, C++, or the like. However, the computer program code for carrying out operations of embodiments of the present invention may also be written in conventional procedural programming languages, such as the "C" programming language or similar programming languages.

Embodiments of the present invention are described above with reference to flowchart illustrations and/or block diagrams of methods, apparatus (systems), and computer program products. It will be understood that each block of the flowchart illustrations and/or block diagrams, and/or combinations of blocks in the flowchart illustrations and/or block diagrams, can be implemented by computer-executable program code portions. These computer-executable program code portions may be provided to a processor of a general purpose computer, special purpose computer, or other programmable data processing apparatus to produce a particular machine, such that the code portions, which execute via the processor of the computer or other programmable data processing apparatus, create mechanisms for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks.

These computer-executable program code portions may also be stored in a computer-readable memory that can direct a computer or other programmable data processing apparatus to function in a particular manner, such that the code portions stored in the computer readable memory produce an article of manufacture including instruction mechanisms which implement the function/act specified in the flowchart and/or block diagram block(s).

The computer-executable program code may also be loaded onto a computer or other programmable data processing apparatus to cause a series of operational steps to be performed on the computer or other programmable apparatus to produce a computer-implemented process such that the code portions which execute on the computer or other programmable apparatus provide steps for implementing the functions/acts specified in the flowchart and/or block diagram block(s). Alternatively, computer program implemented steps or acts may be combined with operator or human implemented steps or acts in order to carry out an embodiment of the invention.

As the phrase is used herein, a processor may be "configured to" perform a certain function in a variety of ways, including, for example, by having one or more general-purpose circuits perform the function by executing particular computer-executable program code embodied in computer-readable medium, and/or by having one or more application-specific circuits perform the function.

Embodiments of the present invention are described above with reference to flowcharts and/or block diagrams. It will be understood that steps of the processes described

12

herein may be performed in orders different than those illustrated in the flowcharts. In other words, the processes represented by the blocks of a flowchart may, in some embodiments, be performed in an order other than the order illustrated, may be combined or divided, or may be performed simultaneously. It will also be understood that the blocks of the block diagrams illustrated, in some embodiments, merely conceptual delineations between systems and one or more of the systems illustrated by a block in the block diagrams may be combined or share hardware and/or software with another one or more of the systems illustrated by a block in the block diagrams. Likewise, a device, system, apparatus, and/or the like may be made up of one or more devices, systems, apparatuses, and/or the like. For example, where a processor is illustrated or described herein, the processor may be made up of a plurality of microprocessors or other processing devices which may or may not be coupled to one another. Likewise, where a memory is illustrated or described herein, the memory may be made up of a plurality of memory devices which may or may not be coupled to one another.

While certain exemplary embodiments have been described and shown in the accompanying drawings, it is to be understood that such embodiments are merely illustrative of, and not restrictive on, the broad invention, and that this invention not be limited to the specific constructions and arrangements shown and described, since various other changes, combinations, omissions, modifications and substitutions, in addition to those set forth in the above paragraphs, are possible. Those skilled in the art will appreciate that various adaptations and modifications of the just described embodiments can be configured without departing from the scope and spirit of the invention. Therefore, it is to be understood that, within the scope of the appended claims, the invention may be practiced other than as specifically described herein.

What is claimed is:

1. A system for implementing authentication control protocols via components of entity device, the system comprising:

a memory device with computer-readable program code stored thereon;

a communication device; and

a processing device operatively coupled to the memory device and the communication device, wherein the processing device is configured to execute the computer-readable program code to:

determine initiation of a resource interaction by a user at an entity device;

control one or more components of the entity device to cause the entity device to capture a real-time image of the user;

retrieve a user image provided by the user a previous time period during processing of one or more resource pools associated with the user;

generate an updated enhanced user image reflecting current appearance of the user for a current time period based on the user image provided at the previous time period, wherein the current time period is a time period associated with the initiation of the resource interaction by the user;

compare the real-time image captured via the entity device with the updated enhanced user image of the user generated for the current time period;

13

in response to comparing the real-time image with the updated enhanced user image, determine if the real-time image matches the updated enhanced user image; and

implement one or more authentication control protocols based on determining if the real-time image matches the updated enhanced user image.

2. The system according to claim 1, wherein the processing device is further configured to execute the computer-readable program code to determine that the real-time image matches the updated enhanced user image and implement the one or more authentication control protocols comprising:

a complete access protocol to allow complete access to one or more resource pools of the user to complete the resource interaction.

3. The system according to claim 1, wherein the processing device is further configured to execute the computer-readable program code to determine that the real-time image does not match the updated enhanced user image and implement the one or more authentication control protocols comprising at least one of:

a limited access protocol to provide limited access to one or more resource pools of the user;

a restricting access protocol to restrict access to the one or more resource pools of the user; and

a blocking protocol to block the resource interaction initiated by the user.

4. The system according to claim 3, wherein the processing device is further configured to execute the computer-readable program code to implement the limited access protocol based on an exposure protocol selected by the user.

5. The system of claim 2, wherein the processing device is further configured to execute the computer-readable program code to perform one or more actions in response to determining that the real-time image does not match the updated enhanced user image, wherein the one or more actions comprise:

increasing an exposure rating associated with the user; transmitting one or more alerts to a register user device of the user;

locking the one or more resource pools of the user based on determining a number of mismatches associated with comparison of the real-time image and the updated enhanced user image is greater than a predetermined limit; and

transmitting an exposure data file of the user to one or more entity partners.

6. The system according to claim 1, wherein the processing device is further configured to execute the computer-readable program code to: modify the updated enhanced user image at periodic intervals.

7. The system according to claim 1, wherein generating the updated enhanced user image comprises:

computing changes associated with progressive appearance of the user between the previous time period and the current time period; and

generating the updated enhanced user image based on the user image provided by the user to reflect the computed changes.

8. A computer program product for implementing authentication control protocols via components of entity device, the computer program product comprising at least one non-transitory computer readable medium having computer-readable program code portions embodied therein, the computer-readable program code portions comprising executable portions for:

14

determining initiation of a resource interaction by a user at an entity device;

controlling one or more components of the entity device to cause the entity device to capture a real-time image of the user;

retrieving a user image provided by the user a previous time period during processing of one or more resource pools associated with the user;

generating an updated enhanced user image reflecting current appearance of the user for a current time period based on the user image provided at the previous time period, wherein the current time period is a time period associated with the initiation of the resource interaction by the user;

comparing the real-time image captured via the entity device with the updated enhanced user image of the user generated for the current time period;

in response to comparing the real-time image with the updated enhanced user image, determining if the real-time image matches the updated enhanced user image; and

implementing one or more authentication control protocols based on determining if the real-time image matches the updated enhanced user image.

9. The computer program product of claim 8, wherein the computer-readable program code portions comprising executable portions for determining that the real-time image matches the updated enhanced user image and implement the one or more authentication control protocols comprising:

a complete access protocol to allow complete access to one or more resource pools of the user to complete the resource interaction.

10. The computer program product of claim 8, wherein the computer-readable program code portions comprising executable portions for determining that the real-time image does not match the updated enhanced user image and implement the one or more authentication control protocols comprising at least one of:

a limited access protocol to provide limited access to one or more resource pools of the user;

a restricting access protocol to restrict access to the one or more resource pools of the user; and

a blocking protocol to block the resource interaction initiated by the user.

11. The computer program product of claim 10, wherein the computer-readable program code portions comprising executable portions for implementing the limited access protocol based on an exposure protocol selected by the user.

12. The computer program product of claim 9, wherein the computer-readable program code portions comprising executable portions for performing one or more actions in response to determining that the real-time image does not match the updated enhanced user image, wherein the one or more actions comprise:

increasing an exposure rating associated with the user; transmitting one or more alerts to a register user device of the user;

locking the one or more resource pools of the user based on determining a number of mismatches associated with comparison of the real-time image and the updated enhanced user image is greater than a predetermined limit; and

transmitting an exposure data file of the user to one or more entity partners.

13. The computer program product of claim 8, wherein the computer-readable program code portions comprising

15

executable portions for modifying the updated enhanced user image is modified at periodic intervals.

14. A computer-implemented method for implementing authentication control protocols via components of entity device, the method comprising:

determining initiation of a resource interaction by a user at an entity device;

controlling one or more components of the entity device to cause the entity device to capture a real-time image of the user;

retrieving a user image provided by the user a previous time period during processing of one or more resource pools associated with the user;

generating an updated enhanced user image reflecting current appearance of the user for a current time period based on the user image provided at the previous time period, wherein the current time period is a time period associated with the initiation of the resource interaction by the user;

comparing the real-time image captured via the entity device with the updated enhanced user image of the user generated for the current time period;

in response to comparing the real-time image with the updated enhanced user image, determining if the real-time image matches the updated enhanced user image; and

implementing one or more authentication control protocols based on determining if the real-time image matches the updated enhanced user image.

15. The computer-implemented method of claim **14**, wherein the method comprises determining that the real-time image matches the updated enhanced user image and implement the one or more authentication control protocols comprising:

a complete access protocol to allow complete access to one or more resource pools of the user to complete the resource interaction.

16. The computer-implemented method of claim **14**, wherein the method comprises determining that the real-time image does not match the updated enhanced user image

16

and implement the one or more authentication control protocols comprising at least one of:

a limited access protocol to provide limited access to one or more resource pools of the user;

a restricting access protocol to restrict access to the one or more resource pools of the user; and

a blocking protocol to block the resource interaction initiated by the user.

17. The computer-implemented method of claim **16**, wherein the method comprises implementing the limited access protocol based on an exposure protocol selected by the user.

18. The computer-implemented method of claim **15**, wherein the method comprises performing one or more actions in response to determining that the real-time image does not match the updated enhanced user image, wherein the one or more actions comprise:

increasing an exposure rating associated with the user; transmitting one or more alerts to a register user device of the user;

locking the one or more resource pools of the user based on determining a number of mismatches associated with comparison of the real-time image and the updated enhanced user image is greater than a predetermined limit; and

transmitting an exposure data file of the user to one or more entity partners.

19. The computer-implemented method of claim **14**, wherein the method comprises modifying the updated enhanced user image at periodic intervals.

20. The computer-implemented method of claim **19**, wherein generating the updated enhanced user image comprises:

computing changes associated with progressive appearance of the user between the previous time period and the current time period; and

generating the updated enhanced user image based on the user image provided by the user to reflect the computed changes.

* * * * *